Conjecture 00000002899: a counterexample to the displayed Kakeya asymptotic lower bound

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We give an explicit family of finite-field Kakeya sets and a Lean-checked refutation of the displayed coefficient-one eventual lower bound in every dimension at least two. The interpretation limit below is part of the result.

1 Original statement and precise scope

The original English statement is:

> Definition: The Kakeya problem over finite fields: the size of sets containing a line in every direction. Conjecture: The dimension spectrum of Kakeya sets: the size of minimal Kakeya sets in d-dimensional space has the explicit asymptotic lower bound of type q^d(1−c/q), attained by tube merging of polynomial methods. (finite Kakeya size asymptotics)

The original Chinese statement is:

> 定义：有限域上的 Kakeya 问题：含每方向线的集合的尺寸。猜想：Kakeya 集的维数谱：d 维空间的最小 Kakeya 集的尺寸有 q^d(1−c/q) 型显式渐近下界且下界由多项式方法的管合并达到。(有限 Kakeya 尺寸渐近)

The source was captured from the repository's conjecture file.

For a finite field F of cardinality q and an integer $d \geq 1$, a Kakeya set $K \subseteq F^d$ contains a full affine line in every direction:

$$\forall v \in F^d \setminus \{0\}, \quad \exists a \in F^d, \quad \forall t \in F, \quad a + tv \in K.$$

The parametrization $t \mapsto a + tv$ is injective when $v \neq 0$: a nonzero coordinate of v permits cancellation. Thus each required line has exactly q points. Rescaling a direction by a nonzero field element only reparametrizes its line, so this definition includes every projective direction. Write $m_d(F)$ for the minimum cardinality of a Kakeya subset of F^d. This minimum exists and is attained because the full space is admissible and the collection of subsets is finite.

We interpret the displayed expression as the eventual lower bound

$$\exists c \in \mathbb{R}\ \exists Q \in \mathbb{N}\ \forall F \text{ finite field}, \qquad q \geq Q \implies m_d(F) \geq q^d(1 - c/q). \tag{1}$$

Both c and Q may depend on the fixed dimension. We disprove (1) for every $d \geq 2$, allowing every real c and every threshold.

Interpretation limit. The source leaves "of type", "dimension spectrum", and "tube merging" undefined. Our result refutes the displayed leading coefficient one with an error of order q^{d-1}. It does not refute an unspecified positive multiple of q^d, such as an $\Omega(q^d)$ bound, or a version allowing c to grow with q. No unspecified multiplicative constant is inserted into the displayed formula. Failure of its necessary lower-bound clause suffices; we do not invent a definition for the additional attainment method or claim to determine the sharp minimum.

2 A family in every dimension at least two

Let F be a finite field with $2 \neq 0$ in F, and let $q = |F|$. Put

$$S = \{z^2 : z \in F\}, \qquad K_d = \{w \in F^d : w_0 = 0 \text{ or } w_1 + w_0^2 \in S\}, \quad d \geq 2.$$

Only the first two coordinates are constrained.

Theorem 1. *The set K_d is Kakeya, and*

$$|K_d| = \frac{q^2 + 2q - 1}{2}\, q^{d-2}. \tag{2}$$

Consequently

$$\frac{m_d(F)}{q^d} \leq \frac{1}{2} + \frac{1}{q} - \frac{1}{2q^2}. \tag{3}$$

Proof. First count the squares, including zero. The fiber of $z \mapsto z^2$ over zero consists of zero alone. For a nonzero square z^2, the fiber is $\{z, -z\}$: equality of squares factors as $(u - z)(u + z) = 0$, and $z \neq -z$ because $2z \neq 0$. Counting the fibers gives

$$q = 1 + 2(|S| - 1), \qquad 2|S| = q + 1.$$

Now fix any nonzero direction $v \in F^d$. If $v_0 = 0$, choose $a = 0$; every point $a + tv$ has first coordinate zero and belongs to K_d. This case also covers directions whose first two coordinates vanish but a later one does not. If $v_0 \neq 0$, set $b = v_1/(2v_0)$, and let $a_1 = b^2$ with every other coordinate of a equal to zero. For every $t \in F$,

$$(a + tv)_1 + (a + tv)_0^2 = b^2 + tv_1 + (tv_0)^2 = (b + tv_0)^2,$$

since $2bv_0 = v_1$. Thus the whole line lies in K_d in this case as well.

For a fixed first coordinate $x = 0$, all q second coordinates are allowed. For each $x \neq 0$, translation $y \mapsto y + x^2$ is a bijection, so exactly $|S|$ second coordinates are allowed. The remaining $d - 2$ coordinates are free. Hence

$$|K_d| = \big(q + (q - 1)|S|\big) q^{d-2} = \frac{q^2 + 2q - 1}{2}\, q^{d-2}.$$

This is a count of the explicitly defined family, not an optimality claim. The genuine minimum satisfies $m_d(F) \leq |K_d|$. Dividing by the positive number q^d proves (3). □

3 Counterexamples beyond every threshold

Theorem 2. *For every integer $d \geq 2$, every $c \in \mathbb{R}$, and every $Q \in \mathbb{N}$, there is an odd prime $p \geq Q$ and a Kakeya set $K \subseteq (\mathbb{Z}/p\mathbb{Z})^d$ such that*

$$m_d(\mathbb{Z}/p\mathbb{Z}) \leq |K| < p^d(1 - c/p).$$

In particular, (1) *is false in every such dimension.*

Proof. Choose a natural number $N \geq Q$ with $N > 2$ and $N > 2(c + 1)$, using the Archimedean property of $\mathbb{R}$. Unboundedness of primes gives a prime $p \geq N$. Thus $p > 2$, so $F = \mathbb{Z}/p\mathbb{Z}$ is an actual odd finite field of cardinality p, and the family of Theorem 1 is available. For this p,

$$p^2(1 - c/p) - \frac{p^2 + 2p - 1}{2} = p\left(\frac{p}{2} - c - 1\right) + \frac{1}{2} > 0.$$

Multiplication by $p^{d-2} > 0$, followed by the minimum comparison, gives the stated strict inequality. The choices were made after arbitrary c and Q; this proves an unbounded family of violations, not a small-field exception. If (1) held, specialize its field quantifier to this prime field to contradict the strict inequality. □

4 Formal correspondence and verification

The Lean project uses `Fin d → F` as its actual coordinate space. `IsKakeya` requires membership of every point on a full affine line for every nonzero vector. `family` constrains coordinates zero and one and leaves all others free. Its explicit subtype equivalence to a first-two- coordinate fiber times `Fin n → F` supplies the higher-dimensional cardinality argument.

`minimumKakeyaSize` is `Nat.find` over cardinalities of all admissible Kakeya finsets, with existence witnessed by the full space. Attainment and comparison with any actual Kakeya finset are separate theorems. The natural counting identity used before casting to $\mathbb{R}$ is

$$2|K_{n+2}| + q^n = (q^2 + 2q)q^n,$$

which avoids truncated subtraction and division in natural numbers.

`ProposedEventualLowerBound` quantifies over arbitrary real c, natural Q, and every finite field. Prime fields suffice to contradict this universal assertion; their unbounded sizes are proved using the generic unbounded-prime theorem. All dimensions $d \geq 2$ are obtained by $d = (d - 2) + 2$.

Checked project and conclusions

The eight source modules are the default root `Conjecture2899`, `Geometry`, `Squares`, `Counting`, `Minimum`, `Bounds`, `Asymptotic`, and `AxiomAudit`; the latter seven are under the `Conjecture2899` namespace directory. The default target reaches them all. The principal declarations in that namespace are

- `counterexamples_in_each_dimension`: Theorem 2, with the actual field, set, minimum comparison, and strict inequality;
- `not_proposedEventualLowerBound`: the explicit negation of (1) for every $d \geq 2$;
- `normalized_minimum_upper_bound_in_each_dimension`: (3) for every finite field with $2 \neq 0$ and every $d \geq 2$.

The global declaration `finite_field_square_count` proves the square fiber count. The complete inventory lists 25 named definitions and theorems.

The project is pinned to Lean 4.19.0 and Mathlib v4.19.0, at commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

All nine dependency revisions are locked. An independent source-only project copy passed the complete Lake build and strict replay of every source module and the Lake configuration with warnings treated as errors. The 11 frozen source and configuration files remained identical. All 39 owned declarations, including 14 generated proof or equation theorems, were audited by originating module. Their axiom dependencies are limited to `propext`, `Classical.choice`, and `Quot.sound`; no unsafe generated helper is present. The audit covers 8,326 reachable declarations and verifies 2,942 imported module identities. All 63 execution receipts were checked independently.

An ordinary safe control was accepted, while five controls containing a custom axiom, a transitively hidden axiom, an admitted proof, compiler-trusted decision, or unsafe definition were rejected as intended. There is no admitted proof, custom axiom, or compiler-trusted decision in the submission. The square count and prime-field counterexamples are symbolic proofs; no external numerical program is used as a proof premise. Separate semantic review checked the actual objects, quantifiers, minimum, and interpretation against both original texts.

Trust and acceptance limits. Generic dependency source identities and cached imported artifacts were checked, but the entire dependency library was not rebuilt and Lean was not bootstrapped. The dependency-graph check is not a second proof kernel; source compilation supplies Lean kernel checking. Runtime binaries and libraries were hashed; imported runtime artifacts were recorded without a compiler bootstrap. PATH safeguards are not operating-system network isolation. The accompanying verification record supplies reproduction commands, raw receipts, independent reviews, and the final PDF inspection. These local checks do not constitute maintainer acceptance or settle alternative interpretations of the informal source.